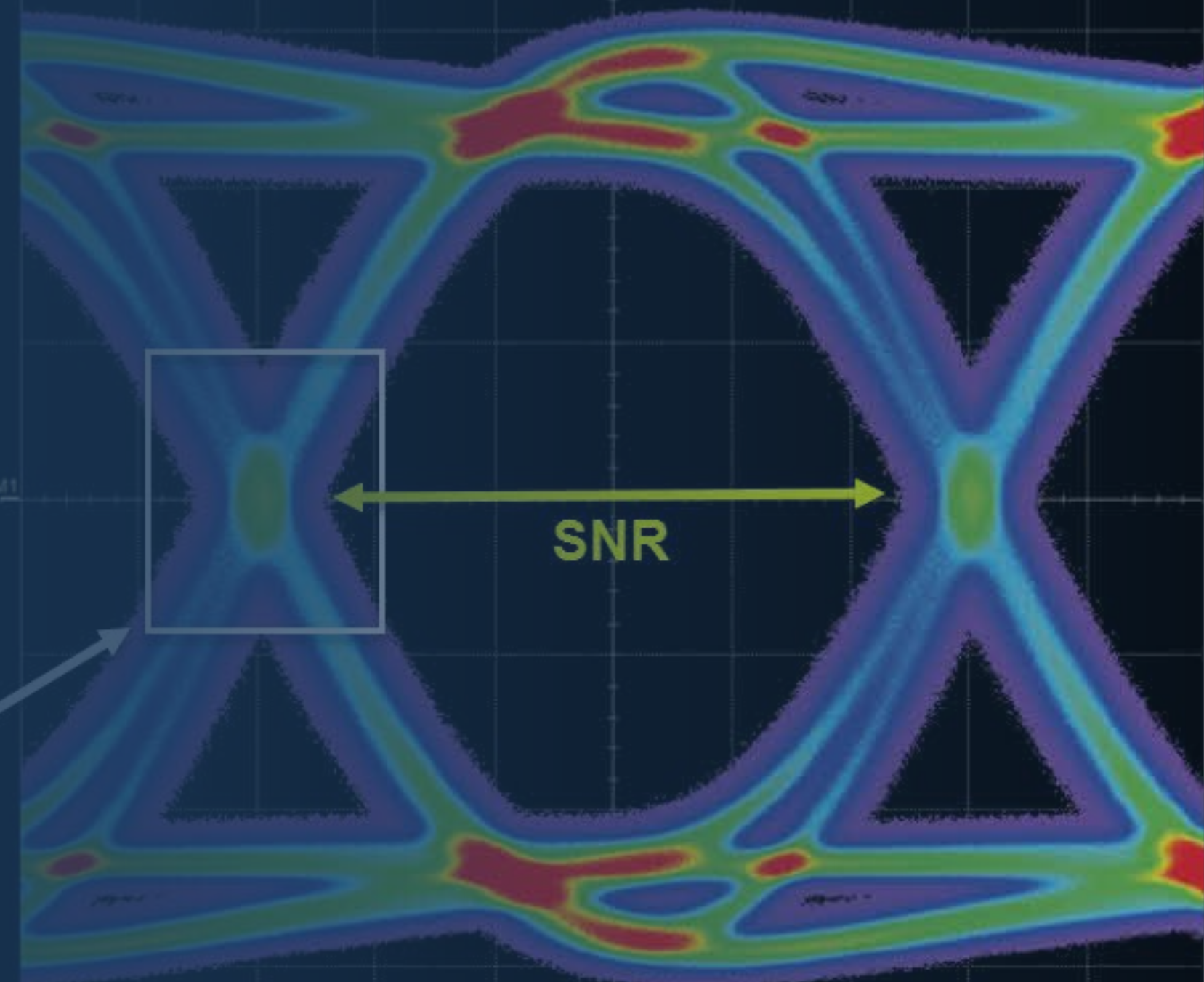


Matched Filter Design & BER Performance Analysis of M-PAM Systems over AWGN Channels

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Communications and Information Engineering

Indication
of Jitter



Timebase	0.0 μ s	Trigger	C2[DC]
	20.0 μ s/div	Stop	0.0 mV
8.00 MS	40.0 GS/s	Edge	Positive

Executive Overview



Theory

Optimal SNR maximization
via the Schwarz Inequality

Deriving the theoretical Q-
function BER limit for M-
PAM signaling



Implementation

MATLAB/Simulink cross-
validation with 30,000
symbols

Independent algorithmic
and architectural
verification

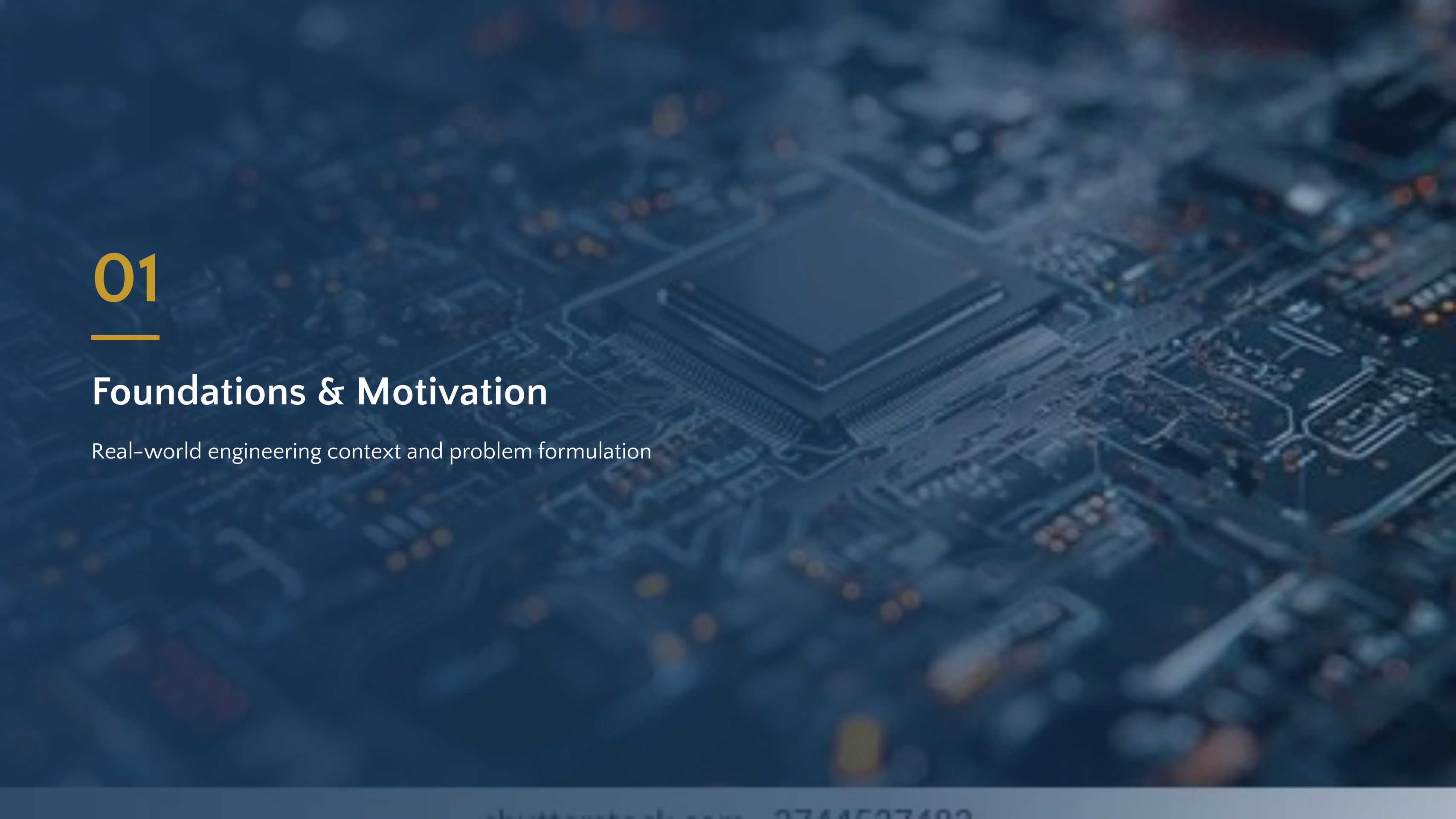


Discovery

Pulse shaping dictates the
success of higher-order
modulation

The 7 dB SNR penalty
tradeoff in 4-PAM systems

Core Insight: We validate the architecture of modern digital links by comparing theoretical limits to simulated reality across multiple pulse shapes and modulation orders.

The background of the slide is a high-resolution, close-up photograph of a printed circuit board (PCB). A large, square integrated circuit (chip) is the central focus, surrounded by a dense network of fine, gold-colored traces and smaller electronic components. The overall color palette is a deep, muted blue, which is a common choice for technical or academic presentations to convey a sense of precision and technology.

01

Foundations & Motivation

Real-world engineering context and problem formulation

Motivation: Real-World Engineering Context



APPLICATION DOMAINS

DSL / Broadband

Multi-level signaling enables high data rates over copper infrastructure

Optical Fiber

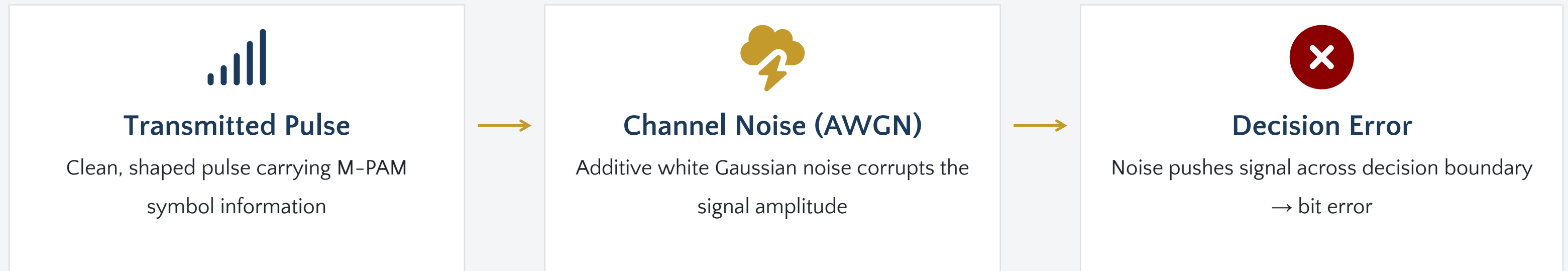
High-speed PAM-4 links power modern data center interconnects

Power-Constrained Reliability

Minimizing error in environments where every dB of SNR counts

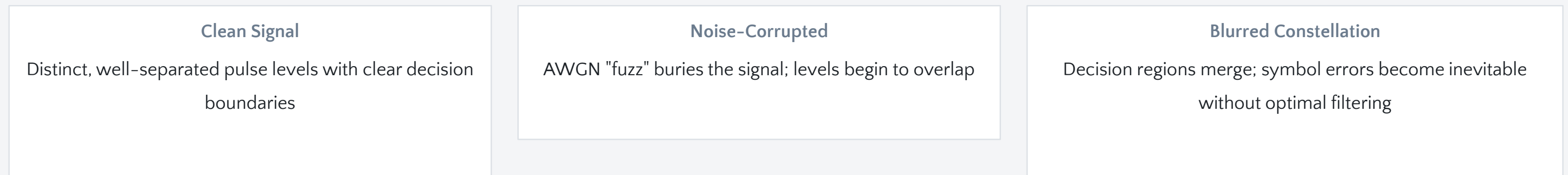
Key Point: M-PAM is the backbone of high-speed backplane signaling. Every decibel of SNR gained through optimal filtering translates directly into higher throughput.

Problem Definition: Signal Corruption in AWGN



The Challenge: How do we extract a clean decision from a signal that is visually indistinguishable from noise?

SIGNAL DEGRADATION VISUAL



Project Objectives: Quantifiable Targets

- 01** **Reach the Theoretical Q-Function Limit**
Design a matched filter that achieves the theoretical Bit Error Rate bound derived from the Q-function for M-PAM systems over AWGN channels.
- 02** **Ensure Statistical Convergence**
Simulate $N = 30,000$ symbols per SNR point to guarantee statistically significant BER estimates, even at low error rates where Monte Carlo variance is highest.
- 03** **Quantify the 4-PAM SNR Penalty**
Measure and validate the theoretical 7 dB SNR penalty when moving from 2-PAM to 4-PAM while maintaining equivalent BER performance.

All targets are verified through dual-platform implementation: MATLAB (algorithmic) and Simulink (architectural)

02

Theoretical Framework

System model, matched filter theory, and error probability

System Model: The M-PAM Chain



TRANSMITTER CHAIN

Bit Source → Random binary data

Level Mapper → Bits to M-PAM levels (Gray coded)

Pulse Shaper → Discrete to continuous-time

CHANNEL

AWGN → Adds white Gaussian noise with variance σ^2

RECEIVER CHAIN

Matched Filter → Maximizes SNR at sampling instant

Sampler → Extracts symbol value at optimal timing

Decoder → Maps levels back to bit decisions

Key Insight: The magic happens between the AWGN channel and the sampler. The Matched Filter is the optimal linear filter for maximizing SNR in the presence of additive noise.

Matched Filter Intuition

DEFINITION

$$h(t) = p(T_s - t)$$

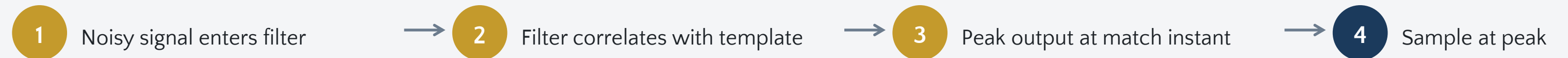
The matched filter impulse response is the time-reversed, delayed version of the transmitted pulse shape

THE INTUITION

"Template Matcher"

The filter looks for its own shape within the noise. When it finds a match, it generates a peak — allowing us to ignore the surrounding noise.

HOW IT WORKS



Result: The Matched Filter maximizes SNR *specifically at the sampling instant*, making it the optimal receiver front-end for AWGN channels. This is proven via the Schwarz Inequality.

Pulse Shape Comparison

Rectangular	Triangular	Raised Cosine ($\alpha = 0.35$)
 Pros: Simplest to implement, easy to generate Cons: High sidelobes, severe ISI in symbol streams	 Pros: Faster decay, reduced sidelobe energy Cons: Moderate ISI still present at sampling instants	 WINNER Pros: Nyquist-optimal, zero ISI at sampling points Result: Best BER performance, ideal for symbol streams

Key Insight: While the Matched Filter is "pulse-blind" for single pulses, the **Raised Cosine** wins in a stream because it prevents symbols from bleeding into each other (ISI).

The Mathematics of Error Probability

THE Q-FUNCTION

$$P_s(e) = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6k}{M^2-1} \cdot \frac{E_b}{N_0}}\right)$$

Symbol error probability for M-PAM in AWGN

INTERPRETATION

The Q-function represents the probability that the noise "vector" is longer than the distance to the nearest decision boundary.

As M increases, the symbol spacing shrinks — hence the error penalty.

VARIABLE DEFINITIONS

Symbol	Definition
M	Modulation order (number of amplitude levels)
k	Bits per symbol: $k = \log_2(M)$
E_b/N_0	Energy per bit to noise power spectral density ratio (dB)
$Q(x)$	Tail probability of the standard normal distribution

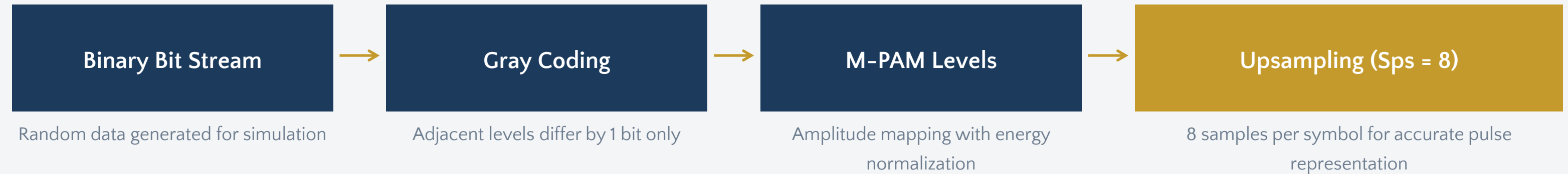
Takeaway: This equation is our theoretical benchmark. Any practical filter design must approach this Q-function limit to be considered optimal.

03

Methodology & Implementation

Transmitter design, channel modeling, and receiver architecture

Transmitter Design: Mapping & Upsampling



WHY GRAY CODING?

In M-PAM, the most likely error is jumping to an **adjacent level**. Gray coding ensures this only results in a **single bit error**, minimizing the BER impact.

WHY Sps = 8?

Using 8 samples per symbol allows accurate representation of the pulse shape's continuous-time properties in a digital system. It provides finer resolution for the Matched Filter peak detection.

Normalization: Scaling factor $d = \sqrt{3/(M^2-1)}$ ensures average energy per symbol = 1, enabling fair comparison across modulation orders.

Channel Modeling: The AWGN Block

NOISE VARIANCE

$$\sigma^2 = \frac{1}{2k(E_b/N_0)}$$

Noise power is inversely proportional to SNR and bits per symbol

KEY PRINCIPLES

1. Unit Energy Normalization — $E_s = 1$ for fair comparison across M
2. No "Cheating" — Gains must come from the receiver, not more power

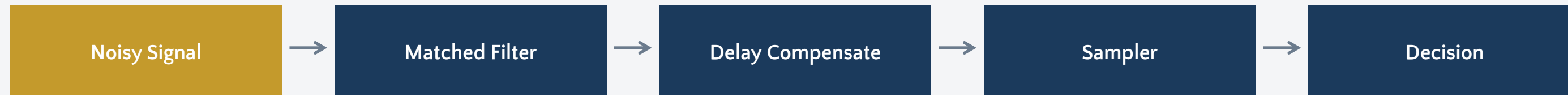
CHANNEL PARAMETERS

Parameter	Value / Formula	Purpose
Noise Variance σ^2	$1 / (2k \cdot E_b/N_0)$	Sets noise power per SNR point
Normalization Factor d	$\sqrt{3 / (M^2 - 1)}$	Ensures unit average symbol energy
SNR Range	0 dB to 14 dB (E_b/N_0)	Covers practical operating range

Critical: Without rigorous energy normalization, 4-PAM would appear to perform better simply because it uses more power — a common pitfall in BER analysis.

Receiver Logic: Delay Compensation

RECEIVER SIGNAL CHAIN



DELAY CALCULATION

$$Delay = 2(L_p - 1)$$

Where L_p = length of the pulse shape filter



Critical Timing

If you sample at the wrong time, you hit the **slope** instead of the **peak**, causing catastrophic BER degradation.

SAMPLING OUTCOMES

✓ Correct Sampling

Sample at the MF output peak → Maximum SNR → Minimum BER

✗ Wrong Sampling

Sample on the slope → Reduced SNR → Severe BER degradation

Simulink Architectural Overview

HARDWARE-IN-THE-LOOP EQUIVALENT

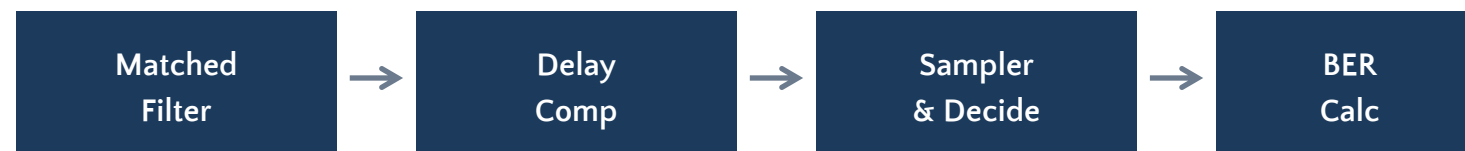
TRANSMITTER



CHANNEL



RECEIVER



MODEL DESCRIPTION

Every block represents a physical component:

- **FIR Filter** — Pulse shaping and matched filtering
- **AWGN Block** — Controlled noise per Eb/NO
- **Decision Threshold** — M-PAM level comparison
- **BER Counter** — Error accumulation (30,000 sym)

Simulink validates that the architectural flow matches algorithmic logic.

Purpose: Simulink provides a "Hardware-in-the-Loop" equivalent, where every block maps to a real physical component in the communication chain.

04

Performance Analysis

BER results, critical comparisons, and cross-validation

2-PAM vs. 4-PAM Tradeoff

MODULATION COMPARISON

Metric	2-PAM	4-PAM
Bits per Symbol	1 bit/symbol	2 bits/symbol
Reliability	High — Wide level spacing	Moderate — Tighter spacing
Throughput	Baseline	2x Data Rate

THE PENALTY

4-PAM requires **7 dB more SNR** than 2-PAM to achieve the same BER — this is the price of doubling the data rate.

WHY 7 DB?

With 4 levels instead of 2, the distance between adjacent symbols shrinks. The Q-function argument scales with the square of this distance. To maintain the same argument (and thus the same BER), the SNR must increase by approximately 7 dB — a fundamental limit of M-PAM modulation.

BER Interpretation: Pulse Shape Impact

✓ Raised Cosine — Tracks Theory

Why it works:

- Nyquist property: zero crossings at all other sampling instants
- No ISI at the decision points
- BER follows the theoretical Q-function limit
- Matched Filter operates at peak efficiency

✗ Rectangular / Triangular — Deviates

Why they fail:

- Non-zero tails at adjacent sampling instants
- Inter-symbol interference (ISI) accumulates
- BER deviates from theory at high SNR
- Matched Filter cannot remove ISI from bad pulses

Key Finding: At high SNR, the gap between Raised Cosine and Triangular pulse BER is **purely due to ISI**. The Matched Filter maximizes SNR against noise, but it cannot fix interference caused by poor pulse shaping.

ISI MECHANISM

In a symbol stream, each pulse overlaps with its neighbors. The Raised Cosine's Nyquist property ensures its tail is exactly zero at every other sampling instant. Rectangular and Triangular pulses have non-zero tails, causing neighboring symbols to "bleed" into the current decision — this is ISI, and it creates an error floor that no amount of SNR can overcome.

MATLAB vs. Simulink Validation



Result: Exact Overlap — The two independent implementations produce identical BER curves across all SNR points and pulse shapes.

MATLAB (Algorithmic)

- Script-based signal processing pipeline
- Direct implementation of mathematical formulas
- Full control over every computation step
- Serves as the **theoretical reference**

Simulink (Architectural)

- Block-diagram based system model
- Each block maps to a physical component
- Validates the architectural design flow
- Serves as the **hardware equivalent**

SIGNIFICANCE

When two independent implementations (algorithmic + architectural) overlap perfectly, the design is **validated**. The minor wiggles at low BER are simply Monte Carlo statistical noise — they diminish with more symbols.



05

Conclusion & Defense

Engineering insights, challenges, and future directions

Engineering Interpretation & Challenges

Challenges Faced

1. Synchronization

Precise symbol timing across 30,000 symbols

2. Es/NO Scaling

Converting Eb/NO and Es/NO for different M

3. Pulse Tail Management

Handling filter transients at stream edges

Engineering Takeaway

Matched filtering is only half the battle.

Pulse shaping is equally critical — a poorly chosen pulse creates ISI that no amount of SNR can overcome.

The combination of **optimal filtering + Nyquist pulse shaping** is what delivers performance that approaches the theoretical limit.

EYE DIAGRAM PRINCIPLE

Open Eye: Wide, clean eye opening → reliable communication → low BER

Closed Eye: Narrow, noisy eye → poor communication → high BER

Conclusion & Future Work



TODAY

Optimized AWGN PAM Receiver



TOMORROW

Next Research Directions

1. **QAM Extension** — Extend the matched filter framework to Quadrature Amplitude Modulation
2. **Adaptive Equalization** — Combat ISI in channels with unknown or varying characteristics
3. **Rayleigh Fading Channels** — Tackle real-world multipath fading beyond the AWGN assumption

Our design is **theoretically optimal** for thermal noise. The next step is tackling the real world: **multipath fading** and **adaptive equalizers**.

Thank You

Technical Discussion & Q&A

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Communications and Information Engineering

Ready for questions regarding the Matched Filter derivation,
ISI analysis, or BER performance evaluation.